

ECEN-601: Homework 3

Due on September 26, 2025 at 11:59 pm

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Problem 1

Prove that if a Cauchy sequence in a metric space (not assumed to be complete) has a limit point, then it has a limit. [A point x is a limit point of $\{x_n\}$ if every neighborhood of x contains infinitely many elements of the sequence.]

Solution

Problem 2

Call two metrics d_1 and d_2 on the same set M equivalent if there exist positive constants c_1, c_2 such that $d_1(x, y) \leq c_2 d_2(x, y)$ and $d_2(x, y) \leq c_1 d_1(x, y)$ for all x and y in M . Prove that $x_n \rightarrow x$ in the d_1 -metric if and only if $x_n \rightarrow x$ in the d_2 -metric, if d_1 and d_2 are equivalent.

Solution

Problem 3

If $f : M \rightarrow N$ and $g : N \rightarrow P$ are continuous, prove $g \circ f : M \rightarrow P$ is continuous. Given an example where $g \circ f$ is continuous but g and f are not.

Solution

Problem 4

Let a be a fixed real number with $1 < a < 3$. Prove that the mapping $f(x) = (x/2) + (a/2x)$ satisfies the hypotheses of the contractive mapping principle on the domain $(1, \infty)$. What is the fixed point?

Solution

Problem 5

If f satisfies the hypotheses of the contractive mapping principle and x_1 is any point in M , show that $d(x_1, x_0) \leq d(x_1, f(x_1))/(1 - r)$ where x_0 is the fixed point. Informally, this says that if $f(x_1)$ is close to x_1 , then x_1 is close to the fixed point (but r must not be too close to 1 for this to be a good estimate).

Solution